

Christopher Mbiba

Economist / Applied Microeconometrics, Bayesian Methods and Machine Learning for Development

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PROFESSIONAL PROFILE

Development economist and MPhil candidate at the University of Cape Coast who studies household welfare, poverty and labour markets in Ghana and Sub-Saharan Africa. Graduated with First Class Honours (CGPA 3.7/4.0) and three Dean's Awards, and has one single-authored paper submitted to the *Journal of African Economies* alongside three co-authored manuscripts under review. Combines causal inference on large household surveys with Bayesian modelling and machine learning, working across GLSS, AHIES, DHS, Afrobarometer and AidData microdata. Builds deployed evidence tools, including a poverty-targeting model (ROC-AUC 0.89) served on Microsoft Azure and an open-source Python package for poverty measurement.

CORE COMPETENCIES AND EXPERTISE

- **Domain expertise:** development microeconomics; poverty, vulnerability and multidimensional deprivation; labour markets and informality; development finance and aid effectiveness; energy poverty; food systems and nutrition; targeting and policy evaluation
- **Econometrics and causal inference:** cross-sectional, panel and time-series econometrics; impact evaluation with propensity score matching, spatial matching and inverse probability weighting; survey-weighted estimation; forecasting
- **Bayesian methods and machine learning:** Bayesian inference and hierarchical modelling; supervised and unsupervised machine learning; model interpretability; causal machine learning
- **Data engineering and deployment:** record linkage and fuzzy matching; reproducible pipelines; API development; containerisation; cloud deployment; CI/CD and automated testing
- **Leadership and operations:** research data management across concurrent projects; training design and delivery; technical reporting to executive bodies; writing for policy and public audiences

PROFESSIONAL EXPERIENCE

Research Assistant (Part-time)

Aug 2025 – Present

School of Economics, University of Cape Coast, Ghana

Supports research led by Dr. Raymond Elikplim Kofinti and Dr. Gloria Essilfie on welfare, poverty, employment and development finance.

- Built analysis-ready datasets from five major sources (GLSS, AHIES, DHS, Afrobarometer and AidData) by cleaning, validating and harmonising raw survey and project files in Stata, R and Python.
- Moved a co-authored study of district poverty in Ghana to journal submission (*Poverty & Public Policy*, October 2026) by specifying its Bayesian network model, producing district-level spatial maps and validating its reference list.
- Kept concurrent projects grounded in current evidence by organising and verifying the literature and turning it into synthesis notes, tables and figures for the lead researchers.

Teaching Assistant (National Service)

Aug 2024 – Aug 2025

Department of Economics, University of Cape Coast, Ghana

- Supported undergraduate learning across four courses by leading tutorials that turned economic theory and statistical methods into accessible, worked examples.
- Kept assessment timely and consistent by marking assignments and examinations and supervising examination sessions for faculty.
- Freed faculty time for research by assisting with research coordination.

PORTFOLIO PROJECTS AND APPLIED SYSTEMS

Poverty Targeting System: ML Prediction API and Dashboard

2026

Stack: Python, scikit-learn, LightGBM, SHAP, FastAPI, Docker, Azure Container Apps, GitHub Actions, Streamlit

Live: poverty-targeting-ghana.streamlit.app | Code: github.com/Mbibachris/poverty-targeting

- Built an end-to-end proxy-means test that identifies multidimensionally poor households from 22 short questionnaire items, using Ghana DHS 2022 microdata.
- Validated the measurement base by reproducing the global MPI from raw survey files; the official UNDP/OPHI headcount for Ghana (24.8%) falls within the estimated bounds of 22.8% to 25.9%.
- Achieved a test ROC-AUC of 0.89, reaching 72% of poor people at a 25% targeting budget versus 25% under random selection, by using cluster-grouped cross-validation, survey weights, a pre-specified selection rule and a sealed test set.
- Prevented a costly deployment error by auditing feature-label circularity and subgroup fairness with cluster-bootstrap intervals, which showed the simpler candidate model missed about 75% of poor childless and small households.
- Shipped the model as a containerised FastAPI service on Microsoft Azure with per-household SHAP explanations, protected by a CI/CD pipeline running linting, 180+ tests and a golden-prediction check before every release.

devecon: Open-Source Python Package for Poverty Measurement

2026

Stack: Python, NumPy, pandas, pytest, GitHub Actions | Code: github.com/Mbibachris/devecon

- Gave researchers a reusable, survey-weight-aware toolkit for Foster-Greer-Thorbecke and Alkire-Foster poverty measures by designing modular identification and aggregation functions.
- Guaranteed reliability across Python 3.9 to 3.12 through an automated test suite with continuous integration; the package now serves as the measurement engine of the Poverty Targeting System.

Reproducible Research Data Pipeline (MPhil Thesis)

2026

Stack: R, fuzzy record linkage

- Linked about 52,000 Afrobarometer Round 9 respondents in 38 countries to AidData GCDF v3.0 project records by building a five-script R pipeline that reconciles place names across administrative boundaries with Jaro-Winkler fuzzy matching and documented quality checks at each stage.

Degrees Without Dividends: Returns to Education in Ghana

2026

Data for Impact Datathon, University of Cape Coast

Methods: Mincerian regression, Oaxaca-Blinder decomposition | Data: GLSS 7 | Code:

github.com/Mbibachris/degrees-without-dividends

- Quantified how returns to schooling differ between Ghana's formal and informal sectors by estimating Mincerian earnings regressions, sheepskin effects and an Oaxaca-Blinder decomposition on GLSS 7 data.

EDUCATION AND ACADEMIC CREDENTIALS

Master of Philosophy (MPhil) in Economics

2025 – 2027 (expected)

University of Cape Coast, Ghana

- Thesis: *Spatial Impact of Chinese Aid on Multidimensional Deprivation in Sub-Saharan Africa*, supervised by Dr. Raymond Elikplim Kofinti, Senior Lecturer, Department of Data Science and Economic Policy.
- Research question: whether household deprivation in Sub-Saharan Africa falls in places that receive Chinese official development finance.
- Data: about 52,000 Afrobarometer Round 9 respondents in 38 countries, linked to geolocated AidData GCDF v3.0 project records through a purpose-built R record-linkage pipeline.
- Identification: propensity score matching and spatial matching, comparing households near Chinese-financed projects with observably similar households elsewhere.
- Concurrently serving as a part-time Research Assistant in the School of Economics and as Chairperson of the GRASAG-UCC ICT Committee.

Bachelor of Arts (Social Sciences), Economics

2020 – 2024

University of Cape Coast, Ghana

- First Class Honours; CGPA 3.7/4.0
- Dean's Award for Academic Excellence, Faculty of Social Sciences (2022/23 and 2023/24) and School of Economics (2023/24)

RESEARCH, PUBLICATIONS AND SCHOLARLY OUTPUT

Single-Authored Paper Under Review

Mbiba, C. (2026). *Not the Crisis but the Recovery: State Dependence in Vulnerable Employment in Ghana*. Submitted to the *Journal of African Economies*, September 2026. Available at SSRN: ssrn.com/abstract=7569681.

- Estimates dynamic random-effects probit models with the Wooldridge initial-conditions correction on three within-year quarterly panels from Ghana's Annual Household Income and Expenditure Survey (AHIES) 2022–2024.
- Finds that, net of worker heterogeneity, last quarter's vulnerable employment raises this quarter's probability of vulnerable employment by 9.5 percentage points in 2022, 16.3 in 2023 and 3.3 in 2024, peaking in the adjustment year rather than the crisis year.

Co-Authored Papers Under Review

Kofinti, R. E., Mbiba, C., & Bosomtwe, K. (2026). *District Multidimensional Poverty in Ghana: Causal Structure Learning with Bayesian Networks*. Under review, *Poverty & Public Policy* (submitted October 2026).

- Role: specified the Bayesian network model, produced the district-level spatial maps and validated references.

Neupane, P., Marasini, A., & Mbiba, C. (2026). *Educated but Excluded: Gendered Education-to-Work Conversion and SDG-Linked Labour-Force Inclusion in Nepal*. Under review, *SAGE Open*.

- Role: conducted the empirical analysis.

Neupane, P., Marasini, A., & Mbiba, C. (2026). *Benchmarking Nepal's Energy Demand Forecasts with Time-Series and Decomposable Models under Structural Change*. Under review, *Heliyon*.

- Role: conducted the forecasting analysis.

Research in Progress

Mbiba, C. *Labour Underutilisation Regimes in Sub-Saharan Africa: An Unsupervised Machine Learning Approach*. Manuscript in preparation; analysis complete.

- Builds a typology of underutilisation regimes from ILO harmonised indicators (LU1–LU4) using K-means and DBSCAN clustering, and flags structurally anomalous countries with anomaly detection.

Mbiba, C. *Calibrated Triggers for Climate-Driven Food Crises: A Decision-Theoretic Bayesian Small-Area Approach to Anticipatory Action in the Horn of Africa*. Analysis stage.

Mbiba, C. *Energy Poverty in Africa*. Working paper; replication code at github.com/Mbibachris/energy_poverty_africa.

- Constructs a Multidimensional Energy Poverty Index (MEPI), identifies its drivers with machine learning, and estimates them with panel econometrics.

GRANTS, FUNDING AND FELLOWSHIPS

Tuition Sponsorship, Seasonal School on Inequalities, Poverty and Deprivation

16–27 Nov 2026

Cluster of Research Excellence in Inequalities, Poverty and Deprivation (CoRE IPD), University of Cape Town, South Africa

- Competitively selected for the two-week seasonal school, with full tuition fees sponsored.

HONOURS, AWARDS AND DISTINCTIONS

First Class Honours, BA Economics, University of Cape Coast

2024

Dean's Award for Academic Excellence, School of Economics, University of Cape Coast

2023/24

Dean's Award for Academic Excellence, Faculty of Social Sciences, University of Cape Coast

2023/24

Dean's Award for Academic Excellence, Faculty of Social Sciences, University of Cape Coast

2022/23

TEACHING, MENTORING AND ACADEMIC SERVICE

Courses Supported as Teaching Assistant

2024 – 2025

Department of Economics, University of Cape Coast

- Principles of Microeconomics and Macroeconomics
- Elements of Microeconomics and Macroeconomics
- Economy of Ghana
- Computer Applications for Financial Economists

Digital Skills for the Modern Postgraduate (Seminar Series)

2026 – Present

GRASAG, University of Cape Coast

- Equipped postgraduate students with research-relevant digital skills by designing and leading an eight-session seminar series.
- Made every session available on demand by building the recording and hosting workflow behind it (OBS Studio, YouTube and Google Sites).

CERTIFICATIONS

IBM Data Science Professional Certificate

Dec 2022 – May 2023

IBM via Coursera

- Completed a ten-course programme in data analysis, SQL, machine learning and visualisation, finishing with a capstone predictive-modelling project in Python and SQL.

MEDIA AND PUBLIC WRITING

Founder and Writer, *The Margin*

Substack economics and policy newsletter

- Makes economic research usable for non-specialists by writing plain-language analyses of economics and policy issues for a general audience.

PROFESSIONAL MEMBERSHIPS AND GOVERNANCE

Chairperson, ICT Committee

Aug 2026 – Present

Graduate Students' Association of Ghana, University of Cape Coast Local (GRASAG-UCC)

- Leads ICT strategy and digital-skills programming for the university's postgraduate community.
- Keeps the Executive Council informed by preparing formal monthly activity reports on the committee's work.

LANGUAGES AND TECHNICAL COMPETENCIES

Languages

- English: native-level proficiency (CEFR C2); spoken from childhood and medium of instruction throughout education
- Konkomba: fluent (spoken)
- Twi (Asante and Akuapem): fluent

Technical Toolkit

- **R:** tidyverse, data.table, haven, brms, Stan, MatchIt, fixest, spdep
- **Python:** pandas, NumPy, scikit-learn, LightGBM, SHAP, PyMC, Matplotlib, Seaborn, FastAPI, Streamlit, pytest
- **Stata:** panel data, discrete-choice models, instrumental variables, survey estimation
- **Bayesian computation:** MCMC with Stan, brms, PyMC and JAGS
- **Data and databases:** SQL, record linkage, web scraping
- **Engineering:** Git and GitHub, Docker, Microsoft Azure Container Apps, GitHub Actions CI/CD
- **Writing and visualisation:** LaTeX/Overleaf, Tableau, Streamlit dashboards

REFERENCES

Available upon request.